THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rectangle is 6 cm long and 8 cm broad. Its diagonal (in cm) is: 1

(a) 11 **(b)** 14 **(c)** 9 **(d)** 10

2. In a right triangle, the hypotenuse is opposite to the angle of: 1

(a) 60° **(b)** 45° **(c)** 90° **(d)** 30°

3. The legs of a right triangle are 5 cm and 12 cm. Its hypotenuse (in cm) is: 1

(a) 12 **(b)** 14 **(c)** 13 **(d)** 17

4. The hypotenuse of a right triangle is 13 cm and one leg is 5 cm. The other leg (in cm) is: 1

(a) 13 **(b)** 11 **(c)** 12 **(d)** 8

5. Which of the following is a Pythagorean triplet? 1

(a) 5, 12, 14 **(b)** 6, 13, 13 **(c)** 5, 13, 13 **(d)** 5, 12, 13

6. The perimeter of a right triangle with legs 5 cm and 12 cm is (in cm): 1

(a) 26 **(b)** 30 **(c)** 31 **(d)** 17

7. A rectangle is 5 cm long and 12 cm broad. Its diagonal (in cm) is: 1

(a) 17 **(b)** 13 **(c)** 12 **(d)** 14

8. In a right triangle with legs a, b and hypotenuse c: 1

(a) $a + b = c$ **(b)** $a^2 - b^2 = c^2$ **(c)** $a^2 + c^2 = b^2$ **(d)** $a^2 + b^2 = c^2$

9. The legs of a right triangle are 9 cm and 12 cm. Its hypotenuse (in cm) is: 1

(a) 16 **(b)** 21 **(c)** 15 **(d)** 14

10. The hypotenuse of a right triangle is 15 cm and one leg is 9 cm. The other leg (in cm) is: 1

(a) 11 **(b)** 13 **(c)** 12 **(d)** 6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In a right triangle the hypotenuse is 25 cm and one leg is 7 cm. Find the other leg. 2

12. Is a triangle with sides 4, 5, 6 cm right-angled? 2

13. The legs of a right triangle are 5 cm and 12 cm. Find its hypotenuse. 2

14. In a right triangle the hypotenuse is 20 cm and one leg is 12 cm. Find the other leg. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 13 m long leans against a wall. Its foot is 5 m from the wall. How high on the wall does it reach? 3

16. Find the perimeter of a right triangle whose legs are 8 cm and 15 cm. 3

17. Rohan walks 9 km east and then 12 km north. How far is he from his starting point? 3

OR

A ladder 15 m long leans against a wall. Its foot is 9 m from the wall. How high on the wall does it reach? 3

18. Find the perimeter of a right triangle whose legs are 7 cm and 24 cm. 3

OR

Rohan walks 8 km east and then 15 km north. How far is he from his starting point? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangular field is 5 m long and 12 m broad. Find the length of its diagonal. A man walks along the diagonal instead of the two sides; how much distance does he save? 5

OR

Using $(2m,\ m^2-1,\ m^2+1)$ with $m = 4$, write a Pythagorean triplet and verify it. 5

20. A rectangular field is 9 m long and 12 m broad. Find the length of its diagonal. A man walks along the diagonal instead of the two sides; how much distance does he save? 5

OR

Using $(2m,\ m^2-1,\ m^2+1)$ with $m = 5$, write a Pythagorean triplet and verify it. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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